



FRAGMENTED TRADE INTELLIGENCE PLATFORM

Growth Command Centre

Technical Architecture & Data Framework v1.4

SEQUENCE

The Inversion → Data Stores → L1→L2 Aggregation → Drift Detection Rules → Packaged Screens → KPIs → Actions

00 The Inversion

How Growth Dashboards Work Today

The standard sequence for sales analytics in Indian distribution — whether built by a Big 4 firm, a systems integrator, or an internal team — runs the same way:

1. **Collect data.** Primary from ERP, secondary from DMS, visits from SFA, scan data from RPA. Weeks spent cleaning and reconciling.
2. **Build a data model.** Star schema. Fact tables for sales, dimensions for geography, product, time. Cloud warehouse or SQL Server.
3. **Design fixed charts.** KPI strip (Primary TY vs LY, SALY%, SALM%). Bar charts by zone and division. Trend lines. Outlet funnel. Outstanding donut.
4. **Add filters.** Date range, RM/RSM/ASM, Region/State/District, Customer, Product. Filters are the only way to change the view.
5. **Ship tabs.** Executive View, Actionables, Analytics Playground, Distributor Drillthrough. Each is a pre-built page. The "Actionables" tab is typically a filtered table with threshold rules: "DB with SALY < 0 = 323."

This is what a mid-size brand's Sales Control Tower looks like. This is what a large FMCG company's Reach Analysis dashboard looks like. Competent tools. Short shelf life.

What Breaks

1. **Data shown, interpretation left to the user.** North-2 shows ₹6.8 Cr at -20% SALY. Why? Outlet churn? Category mix? Competitive entry? The user cross-references tabs and applies tribal knowledge to form a hypothesis. The dashboard surfaces the symptom. Diagnosis is manual.
2. **Actionables are filtered dumps.** 323 distributors with negative SALY, listed flat. No prioritisation by magnitude. No spatial clustering. No causal attribution. The user sees the count and thinks: now what?
3. **Connected signals are visually disconnected.** Sec: Pri dropped from 87% to 69% in three months. Primary sales show -11% SALY. These are causally linked — pipeline stuffing. But they sit on different panels with no connection drawn.
4. **Reach analysis is a snapshot.** GEO ECO drops from 6.24L to 3.77L in one month. The cliff shows on the trend chart. But which town classes? Which regions? Data quality issue or real churn? The user manually scans tables. The Untapped Potential scatter (Market Share × ND%) shows the opportunity space but doesn't rank, cluster, or route it to anyone.
5. **Observation and action are separate systems.** The user sees a problem, then picks up the phone, opens WhatsApp, sends a voice note, or emails a request. The dashboard ends at observation. The conversation, the context, the follow-up — all happen outside.

The Inversion

The Command Centre reverses the sequence:

LAYER	TRADITIONAL BI	COMMAND CENTRE
Data	Star schema per project. Schema change = rebuild.	Canonical ontology (C01). Fixed schema, Semantic Matcher handles variability. Data cleaning is a platform capability.
Aggregation	Live queries at render time.	Pre-computed L2 cubes on data refresh. Screens read cached aggregates.
Interpretation	Left to the user.	Drift detection after every L2 refresh. Causal decomposition automated. Findings ranked by severity.
Presentation	Fixed charts, filter-driven.	Deterministic screens, scope-driven. Findings feed replaces "check every tab."
Action	Ends at observation.	Two action types: Discussion Actions (80%) with shareable findings for WhatsApp/calls, and War Room Actions (20%) that hand off to modules for half-yearly design exercises.

What This Looks Like in Practice

North-2 at ₹6.8 Cr, -20% SALY. In the traditional dashboard, a red bar. In the Command Centre:

The drift engine already caught it. DA-01 and DA-02 ran on the latest L2 refresh. The finding is on the landing screen: "North-2 secondary declined 20% vs SPLY. Outlet churn in 3 districts accounts for 14pp. Extraction rate at surviving outlets declined 6pp, driven by a declining product category." The RSM opens the finding, sees the spatial decomposition, and picks up the phone to call the ASMs in the affected districts. The finding card — with the map, the decomposition, the distributor list — is shareable as an image on WhatsApp. The ASM responds with a voice note adding ground context: "two distributors switched to competitor last month, a third has a credit dispute." That context gets tagged back to the finding as a note in the org context graph.

The Sec:Pri drop from 87% to 69%. DA-03 fires. The finding shows which distributors are driving the divergence and whether the timing correlates with scheme loading. The RSM shares this on the regional WhatsApp group with the question "is this intended loading or stuffing?" Responses come back with context. The finding accumulates notes.

The GEO ECO cliff from 6.24L to 3.77L. DA-08 fires. The finding decomposes by town class and region, cross-checks against SFA data freshness to distinguish real churn from data lag. The user does not scan tables — the decomposition is already done.

The Untapped Potential scatter. The system already computed the under-penetrated, under-indexed towns in the L2 demand-vs-reach gap. Ranked them. Clustered adjacent ones into expansion zones. This is a standing list, not a finding that fires — it's always available on S-09. When it becomes actionable (budget approved, expansion cycle), it feeds a war room exercise.

00A Incumbent Architecture vs Proposed

How a Typical Sales Control Tower Is Built

Based on observed implementations across Indian FMCG, footwear, and consumer goods companies, the standard Sales Control Tower follows this architecture:

LAYER	TYPICAL IMPLEMENTATION
Source Systems	S4HANA (ERP), Salesforce/SFA (orders, visits, beat data, outlet master, hierarchy), SFTP (scan data, redemption files), Outlook (manual attachments for outstanding, collection). 4–6 source systems.
ETL	15–25 Python scripts and SQL queries. Scheduled daily (7–8:30 AM) via a job scheduler (JS7 or similar). Each job pulls from one source, transforms, and loads into a staging database. Average execution: 30 seconds to 35 minutes per job. Manual jobs for outstanding and collection data (Outlook attachments). Success mailers on completion.
Staging Database	SQL Server. One staging schema with 20–30 tables. Views built on top for the BI layer (vw_l1_secondary_visit_order, vw_primary_invoice_data, etc.). Stored procedures for complex transformations. No spatial indexing.
BI Layer	Power BI. Mix of Native Query (direct SQL) and Power Query (in-memory transform) connections. 20–25 data tables loaded into the Power BI model. Scheduled refresh 1–2x daily. Row-level security by role (RM, RSM, ASM) mapped to geography.
Presentation	5–6 pre-built tabs: Executive View, Actionables, Analytics Playground, Distributor Drillthrough, Report View, Master Download. ~80 KPIs across all screens. Filters: date range, RM/RSM/ASM, region/state/district, customer, product hierarchy.
Access Control	Power BI row-level security. 20–40 named users. Role mapping by region and portfolio (Integrated vs category-specific). Manual user provisioning.
Support	2-person team: one for dashboard development, one for data layer. All changes go through the consulting team. No self-serve capability for the client.

What This Architecture Cannot Do

- **No spatial intelligence.** Outlets have lat/lng but there is no spatial indexing, no micromarket assignment, no coverage geometry, no H3 grid. Geography is flat text fields (state, district) not shapes.
- **No causal decomposition.** The "Actionables" tab applies threshold filters (SALY < 0, Sec:Pri < 50%) but does not decompose what is driving the breach or cluster related breaches spatially.

- **No drift detection.** Thresholds are hardcoded and checked at render time. There is no background process comparing current vs historical patterns, detecting structural shifts, or ranking findings by severity.
- **No ontology.** Schema is designed per project. If the same team moves to a second client, they rebuild the staging tables, views, and ETL jobs from scratch. Nothing is reusable except the Power BI templates.
- **No action path.** The dashboard ends at observation. There is no mechanism to route a finding to a module, share it contextually on WhatsApp, or track what happened after someone saw the number.
- **Manual data dependencies.** Outstanding and collection data arrives via Outlook attachments and is loaded manually. SFA scan data arrives via SFTP with manual triggers. These gaps create data freshness inconsistencies across KPIs.

Proposed Architecture: Clarynt Growth Command Centre

The proposed architecture replaces the per-project staging + Power BI pattern with a platform that is reusable across clients and verticals. The key structural differences:

COMPONENT	INCUMBENT (PER-PROJECT)	PROPOSED (PLATFORM)
Ingestion	Custom Python scripts per source. 15–25 jobs. Each client is a new set of scripts.	Semantic Matcher (4-pass) maps any client schema to canonical L1. New client = new mapping dictionary, same pipeline.
Data Model	Custom staging tables per client. Views per dashboard. No entity model.	Canonical entity model (C01): outlet, distributor, salesperson, SKU, territory, micromarket. Fixed across verticals.
Spatial	None. Lat/lng stored but not indexed. Geography = text.	H3 spatial indexing (res 7–9). Every entity assigned to micromarket. Coverage geometry computed. Choropleth, hexagon, point cluster rendering.
Aggregation	Power BI in-memory model. Live queries at render. Slow at scale.	Pre-computed L2 cubes (DuckDB/Parquet). Materialised on data refresh. Screens read cached aggregates.
Intelligence	Hardcoded threshold filters. No background processing.	Drift detection engine: 10+ rules running post-aggregation. Causal decomposition. Severity ranking. Findings persisted.
Presentation	Power BI. Pre-built tabs. Filter-driven.	React frontend. Deterministic screens with role-based assembly. Findings feed on landing. Contextual spatial panels. State persistence.
Action	None.	Discussion actions (shareable cards, WhatsApp, org context graph). War room actions (module/agent handoff for strategic exercises).
Reusability	Near zero. Each client is a rebuild.	95–100% engine reuse. Only data definitions and terminology change per vertical.

The Compounding Loop: Context Graphs

Traditional BI captures state — what changed. SFA and DMS record that secondary sales dropped, that a distributor stopped billing, that visit compliance fell. But there is no learning loop. The system records what happened and shows it on a chart. Next month, it records again. Nothing compounds.

The Command Centre operates on a different loop — one that captures reasoning, not just state:

6. **Drift detected.** L2 aggregation + drift rules identify what moved and decompose why (causal attribution).
7. **Decision trace captured.** The RSM sees the finding, shares it, discusses it. The ASM responds with ground context. Notes, voice transcripts, photos, location markers are tagged back to the finding. The decision — what was done and why — is recorded.
8. **Outcome joined.** Next data refresh shows whether the metric improved, held, or worsened. The system links the decision trace to the outcome. Did the action work?
9. **Agents get smarter.** Over time, the context graph accumulates: which types of drifts in which geographies responded to which interventions. Drift rules can be re-weighted. War room exercises can be prioritised by historical response patterns. The org context graph becomes the compounding asset.
10. **Better decisions.** The next time a similar drift surfaces, the finding card carries not just the decomposition but the historical context: "Last time this pattern appeared in a similar geography, the team did X and it resolved in 6 weeks."

This is the B2B equivalent of the recommendation engine loop in consumer platforms — but instead of capturing behaviour (what you clicked), it captures reasoning (why you decided). Instead of serving recommendations, it serves findings with historical context. The loop compounds with every engagement cycle.

User Journey: New Client Onboarding

The onboarding path from raw data to a live Command Centre:

11. **Data Collection.** Standard data request template: primary sales (distributor × SKU × date), secondary sales (outlet × SKU × date), outlet master (with lat/long), distributor master, sales hierarchy, beat/territory structure, product hierarchy, outstanding/collection, SFA visit logs, scheme/promo data.
12. **Semantic Mapping.** 4-pass Semantic Matcher maps client columns to canonical L1 schema. 70–80% auto-mapped, 20–30% human-confirmed. Mapping dictionary saved for future refreshes.
13. **Entity Resolution & Geo.** Dedup, geo-normalise, H3 assign, micromarket place, enrich (73 checks).
14. **L1 Verification.** Solutions team + client confirm entity counts, dedup rates, geo-validation results. L1 locked.
15. **L2 Aggregation.** 9 cube tables materialised. Baselines established. 12 months history preferred.

16. **Drift Configuration.** Default thresholds loaded and reviewed with client. Adjusted per category/industry.
17. **Screen & Role Setup.** Role mapping, default scope, user provisioning, screen state initialised.
18. **Go Live.** First findings surface. Walkthrough: drill-down, sharing, context tagging.
19. **Compounding begins.** Each refresh → L1 update → L2 re-aggregate → drift re-run → new findings. Each finding discussed → context tagged → outcome joined. The loop starts turning.

01 Data Stores

Three logical stores. L0 is raw client data. L1 is the ontology-mapped, enriched, georeferenced data that Module 01 produces. L2 is the aggregation layer that the Command Centre reads from. The Command Centre never touches L0 or L1 directly.

L0: Raw Ingestion Store

What comes in from the client. ERP extracts, SFA dumps, DMS exports, flat files. Untouched. Stored as-is with ingestion timestamp and source provenance. This is Module 01’s input.

Store: S3 / local filesystem (engagement-level partitions)
Format: CSV, Excel, JSON, Parquet
Retention: permanent (audit trail)
Access: Module 01 AI Onboarding Agent only

L1: Canonical Object Store

What Module 01 produces after semantic matching (4-pass: hard code, semantic similarity, regex, LLM), entity resolution, geo-normalisation, micromarket assignment, and the 73-check enrichment pipeline. Every record mapped to C01 canonical entity model and spatially indexed.

L1 Entity Tables

ENTITY TABLE	PRIMARY KEY	SPATIAL BINDING
outlet_master	outlet_id (deduplicated, entity-resolved)	lat, lng, h3_index (res 7–9), pincode, micromarket_id, district_id, state_id
distributor_master	distributor_id	depot lat/lng, h3_index, coverage_geometry (convex hull of served outlets)
salesperson_master	sp_id	assigned_beat_ids[], home_pincode
sku_master	sku_id	None (maps through outlet_sku_sales)
territory_master	territory_id (beat → area → region → zone)	beat_geometry (polygon), area/region/zone from beat unions
micromarket_ref	mm_id (Clarynt reference grid)	h3_index, pincode, boundary_geometry, centroid, district_id, state_id

L1 Transaction Tables

TRANSACTION TABLE	GRAIN	TEMPORAL KEY
primary_sales	distributor_id × sku_id	invoice_date (daily grain)
secondary_sales	outlet_id × sku_id	bill_date (daily grain)
outlet_visit_log	sp_id × outlet_id	visit_date, check_in_time, check_out_time
scheme_transactions	scheme_id × outlet_id	transaction_date, scheme_start, scheme_end
scan_data (RPA)	outlet_id × sku_id	scan_date (daily grain, from retailer POS/scan)
competitor_signals	outlet_id × source	observation_date

L1 Reference & Enrichment Tables

REFERENCE TABLE	CONTENT
geo_hierarchy	Country → State → District → Sub-district → City/Town → PIN code. Shapefile-backed. Town class (Metro, TLP, FLP, OLP, OLM-50k-P, OLM-25k-50k, OLM-25k-M). Village priority by population class (25k+, 10-25k, 5k-10k, <5k).
demographic_layer	Population, households, RWI, night light, land cover, construction activity per PIN code / H3 cell.
poi_layer	Cross-industry POI: Clarynt reference universe of viable outlets per micromarket. "Ideal outlet count as per DPL" equivalent.
benchmark_layer	Sector benchmarks from Nielsen, Kantar, IQVIA, syndicate sources. Source provenance tracked.
category_template	Vertical-specific variable trees, consumption norms, outlet type definitions.

Store: DuckDB + h3-duckdb (analytical) + RDS Postgres (state, metadata)

Spatial index: H3 res 7 (primary), 8–9 for urban density

Partitioning: client_id → engagement_id → data_refresh_date

02 L1 → L2 Aggregation

L2 is a set of pre-computed spatio-temporal cubes on top of L1. Every number has a where, a when, and a causal dimension. Aggregation runs on data refresh, not on screen load.

Aggregation Dimensions

- Spatial:** outlet → beat → micromarket (H3) → pincode → town/village → sub-district → district → state → zone → national. Roll-up and drill-down. Town class and village population class are cross-cut attributes (not hierarchy levels).

- **Temporal:** day → week → month → quarter → year. Trailing: L30D, L90D, MTD, QTD, YTD. Comparison: SPLY (same period last year), SALM (same as last month), previous period. MTD completion % (days elapsed / total days in month — the 58%/42% bar in the reference dashboards).
- **Causal:** channel (GT/MT/QC/ecommm) × category/division × outlet_class × territory_owner (ASM/RSM) × distributor. These are the decomposition dimensions for every drift.

L2 Aggregate Tables

L2 TABLE	GRAIN	METRICS	SOURCE
l2_reach	mm_id × channel × town_class × period	active_outlets, total_viable (POI), ND%, geo_tagged_count, geo_eco_count, geo_eco_%, outlet_additions, outlet_churn, coverage_gap	outlet_master + poi_layer + secondary_sales. C05.
l2_extraction	mm_id × channel × category × period	secondary_value, secondary_volume, avg_sales_per_outlet (WSP), WD%, extraction_rate, benchmark_extraction, extraction_gap, UoS (unique outlets selling), lines_per_call	secondary_sales + scan_data + benchmark. C05.
l2_pipeline	distributor_id × category × period	primary_value, primary_volume, secondary_value, secondary_volume, dms_value, pipeline_ratio (sec:pri), billing_frequency, days_stock, OFR (order fill rate), SALY_primary, SALY_secondary	primary_sales + secondary_sales + scan_data.
l2_channel_mix	mm_id × period	gt_share, mt_share, qc_share, ecomm_share, gt_share_delta_sply, structural_shift_flag	Channel Share Engine (M06).
l2_demand	mm_id × category × period	demand_index, demand_rank, demand_vs_distribution_gap, hotspot_score, market_share_estimate, urban_town_potential (GSV)	Demand Disaggregation C09. Hotspot Builder (M03).
l2_territory	beat_id × period	planned_outlets, actual_visited, visit_compliance_%, productive_calls_%, MAU (monthly active users), PJP_MAU, QAU (quarterly active), avg_lines_per_call, beat_revenue, beat_revenue_vs_design	outlet_visit_log + secondary_sales + territory_master.
l2_promo	scheme_id × mm_id × period	baseline_value, actual_value, uplift_value, uplift_%, roi, eligible_outlets, participating_outlets, participation_rate	scheme_transactions + secondary_sales. M05.

l2_outstanding	distributor_id × period	total_outstanding, outstanding_>30d, outstanding_<30d, collection_value, collection_rate, outstanding_vs_primary_ratio	ERP/accounts receivable + primary_sales.
l2_feature_store	mm_id (snapshot)	population, households, RWI, night_light, land_cover, outlet_density, poi_density, prosperity_index, infra_count, construction_activity	demographic_layer + poi_layer. Feature Store C08.

Aggregation Execution

20. Data refresh event fires (new sales file, SFA sync, DMS push)
21. L1 transaction tables updated
22. L2 aggregation jobs run in dependency order via Celery task chain
23. Drift detection rules run against fresh L2 (Section 03)
24. Drift results written to drift_findings in Postgres
25. Command Centre reads L2 + drift_findings on next load

03 Drift Detection Rules

This is the engine that earns the Command Centre its right to exist. Drift detection runs after every L2 aggregation. Every finding has five properties:

- **Where:** spatial entity where the drift was detected.
- **When:** time window (single period or multi-period trend).
- **What:** which L2 metric drifted and by how much.
- **Why:** causal attribution — which dimensions (channel, category, outlet class, territory owner, distributor) explain the drift. Dimensional contribution analysis decomposes the total drift.
- **So what:** Category A → drill-down tactical review on a packaged screen. Category B → action handoff to a Clarynt module or Growth Agent.

Category A: Drill-Down Tactical Review

User sees the finding. Drills into a packaged screen. Investigates. Takes manual action or dismisses.

ID	RULE	LOGIC	THRESHOLD	DRILL - DOWN
DA-01	Reach erosion	ND% at mm × channel drops vs SPLY by >threshold for N consecutive periods.	>3pp sustained 2+ months	S-01 Reach Health

DA-02	Extraction decay	WSP (avg sales/outlet) declining vs benchmark. Extraction rate below category median.	>15% below benchmark 2+ months	S-02 Extraction Health
DA-03	Pipeline stuffing	Sec:Pri ratio below threshold. Primary rising, secondary flat/declining. Cross-check against active scheme loading windows.	Sec:Pri <60% for 2+ months	S-03 Pipeline Health
DA-04	Channel structural shift	GT share at mm declining 3+ consecutive quarters. Structural vs seasonal test.	>2pp/qtr sustained 3+ qtrs	S-04 Channel Mix
DA-05	Beat productivity drift	Beat revenue vs design intent drifting. Visit compliance or MAU dropping. Productive calls declining.	>20% below design 2+ months	S-05 Territory Health
DA-06	Promo underperformance	In-flight uplift below baseline at scheme midpoint.	<50% target uplift at midpoint	S-06 Promo Health
DA-07	Benchmark gap widening	Client extraction or reach vs sector cohort median widening N consecutive months.	Widening 3+ months	S-07 Benchmark
DA-08	Outlet churn spike	Churn (outlets not billing 2+ months) exceeding historical norm at mm/district. Decompose by town class and village priority class.	>2x historical monthly churn	S-01 Reach Health
DA-09	Outstanding concentration	Outstanding >30 days growing at distributor level while collection rate declining. Outstanding:Primary ratio exceeding norm.	>30d outstanding >30% of primary	S-08 Outstanding Health
DA-10	SFA adoption decay	MAU, PJP MAU, or QAU declining vs trailing 3-month average. OFR dropping. Active users as % of total users declining.	MAU decline >10% vs L3M avg	S-05 Territory Health

Category B: War Room / Strategic Action

These drifts accumulate evidence for a strategic exercise. They don't trigger instant module execution — they build the case. When enough Category B findings cluster in a geography or theme, they form the brief for a war room exercise.

ID	RULE	TRIGGER	WAR ROOM ACTION	MODULE / AGENT
DB-01	Whitespace opportunity	Demand index >P75 but ND% <20%. "Under-penetrated towns" from untapped potential analysis. Persistent 2+ periods.	Run expansion targeting. Generate target outlet list + SD network additions estimate.	M02 Gap Scoring + M04 Network Design
DB-02	Under-indexed markets	Market share < ASM market share at town level. ND% >60% but extraction below category benchmark. "Present but under-performing."	Run extraction diagnostic. Identify SKU gaps, visit frequency issues, scheme coverage gaps.	M02 Extraction Benchmarking + M05 Scheme Design
DB-03	SD rebalance needed	Coverage radius exceeding threshold. Beat productivity diverging across SDs under same distributor.	Run SD appointment agent with updated coverage parameters.	Rural Growth Agent 01
DB-04	Route re-optimisation	Visit compliance below threshold AND productive calls declining. Beat geometry mismatched to outlet density.	Run beat re-routing with current outlet universe.	Rural Growth Agent 02 / MR Routing (Pharma)
DB-05	Promo course-correction	DA-06 fires AND participation rate below threshold.	Mid-course scheme adjustment: revise eligibility, adjust parameters.	M05 Scheme Design Engine
DB-06	Channel strategy revision	DA-04 fires for cluster of adjacent micromarkets.	Run archetype reclassification + revised channel playbook.	M06 Archetype + Playbook Engine
DB-07	Demand model refresh	Actual secondary deviating >30% from model prediction for 3+ months.	Flag model staleness. Trigger re-calibration.	M03 Demand Forecast Engine

Drift Finding Schema

Table: drift_findings (RDS Postgres)

finding_id, rule_id, category (A|B), severity (critical|warning|info),
geo_level, geo_id, geo_ancestors (jsonb),
metric_name, current_value, comparison_value, drift_magnitude, drift_direction,
causal_dims (jsonb: channel, category, outlet_class, territory_owner, distributor),
period_start, period_end, consecutive_periods,
summary_text (auto-generated one-liner),

target_screen (Cat A), target_action (Cat B: {module, agent, action_type, params}),
status (new|viewed|actioned|dismissed), created_at, actioned_at

04 Packaged Screens

Screens are deterministic. Fixed structure. Role-based visibility. Configurable scope with state maintenance. Users can pin additional metrics but cannot remove defaults. Contextual detail panel on click shows spatio-temporal-causal decomposition of any metric.

Screen Design Principles

Each screen follows the same structure, informed by what works in the reference dashboards and what fails:

- 26. **KPI strip at top:** 4–6 KPIs with current value, comparison (SPLY, SALM, benchmark), delta, and MTD completion context. This is the one thing traditional dashboards do well — keep it.
- 27. **Drift findings relevant to this screen:** Below the KPI strip, any active findings from Section 03 that route to this screen are displayed as cards with severity badges. This is what the traditional "Actionables" tab tries to be, but with causal attribution and spatial context instead of flat threshold tables.
- 28. **Primary decomposition view:** The main analytical component. A table, chart, or map that decomposes the screen’s topic by the spatial and causal dimensions. This is deterministic — same chart type every time. What changes is the data scope.
- 29. **Spatial panel (contextual):** Appears when the screen’s data has geographic variation. Choropleth, H3 hexagons, or point clusters depending on the metric.
- 30. **Trend panel:** Trailing 6–12 months for the key metric. This is where pipeline ratios, reach trends, and extraction trends live.

Screen Catalogue

ID	SCREEN	COMPONENTS	DRIFT RULES	EQUIVALENT IN REFERENCE
S-00	Landing	Ranked drift findings feed (all Cat A + B). Summary KPI strip: total revenue, ND%, WD%, pipeline ratio, collection rate vs SPLY. No charts — just findings and topline numbers. MTD completion bar.	All	Replaces: Executive View top strip + Actionables sidebar
S-01	Reach Health	KPIs: ND%, GEO ECO, GEO ECO%, outlet count, additions, churn, coverage gap. Table: mm-level reach by town class with threshold highlighting. Choropleth: ND% or GEO ECO% by mm. Trend: ND%	DA-01, DA-08	Replaces: Reach Analysis tab + outlet funnel

		12mo. Outlet funnel (geo-tagged → PJP → billed → active).	
S-02	Extraction Health	KPIs: WD%, WSP/outlet, extraction rate, UoS, lines/call. Table: mm × category extraction vs benchmark. Bar: extraction distribution. Choropleth: extraction gap by mm.	DA-02 Replaces: manual cross-tab of secondary sales vs benchmark
S-03	Pipeline Health	KPIs: Sec:Pri ratio, DMS:Pri, Scan:Sec, days stock, OFR, billing frequency, SALY% primary, SALY% secondary. Table: distributor-level pipeline. Trend: Sec:Pri 6mo. Scatter: primary vs secondary by distributor (stuffing outlier detection). Cross-reference active schemes timeline.	DA-03, DA-09 Replaces: Sales Ratio panel + outstanding donut
S-04	Channel Mix	KPIs: GT/MT/QC/ecom share, GT share delta. Stacked bar: channel share by mm. Trend: share 12mo. Archetype classification table.	DA-04 Not present in reference dashboards (gap)
S-05	Territory & SFA Health	KPIs: visit compliance, productive calls, MAU, PJP MAU, QAU, OFR, active users %. Table: beat-level breakdown. Map: beat boundaries with productivity heatmap. Trend: MAU/UoS 6mo.	DA-05, DA-10 Replaces: RPA/scan tab + Secondary KPIs panel
S-06	Promo Health	KPIs: active schemes, total spend, avg uplift, avg ROI. Table: scheme-level performance. Waterfall: baseline → uplift. Choropleth: participation rate.	DA-06 Not present in reference dashboards (gap)
S-07	Benchmark	KPIs: gap to median, gap to P75, percentile rank. Bar: client vs cohort by geography. Trend: gap 12mo.	DA-07 Not present in reference dashboards (gap)
S-08	Outstanding Health	KPIs: total outstanding, >30d outstanding, collection rate, outstanding:primary ratio. Table: distributor-level aging. Trend: outstanding vs collection vs primary 6mo (the triple-line chart from reference).	DA-09 Replaces: Outstanding chart
S-09	Untapped Potential	Table: state × total towns × under-penetrated × under-indexed. Scatter: Market Share × ND% by town/mm (from reference). Detailed view: town-level with ideal outlets, total outlets, ND%, sales, market share, ASM market share, infra count. Active DB-01 and DB-02 findings highlighted.	DB-01, DB-02 Replaces: Untapped Potential tab

Role-Based Defaults & State

Every role sees the same screen templates but with different default scope (NSM: national, RSM: region, ASM: area) and different screen visibility. Screen state (scope, pinned metrics, sort, last accessed) persists per user in Postgres. Scope overrides are sticky across sessions.

Contextual Detail Panel

Click any metric on any screen → slide-out panel showing: the spatio-temporal decomposition (which geographies contributed, which periods), the causal decomposition (which dimensions drove the movement), and any drift findings linked to that metric. This is the path from "seeing a number" to "understanding why" without switching tabs or cross-referencing filters.

05 KPI Framework

Every KPI is computed from L2 aggregates, carries spatial-temporal-causal context, and is linked to drift rules that monitor it. The KPI causality chain enables decomposition: when revenue drifts, the system walks the chain to identify whether it is a reach problem, extraction problem, pipeline problem, or mix/pricing problem.

Causality Chain

Revenue = $f(\text{Reach} \times \text{Extraction} \times \text{SKU depth} \times \text{Pricing})$

Reach = $f(\text{Outlet count} \times \text{Coverage geometry} \times \text{Visit compliance} \times \text{Town class penetration})$

Extraction = $f(\text{WSP/outlet} \times \text{Category depth} \times \text{Lines/call} \times \text{Scheme effectiveness})$

Pipeline = $f(\text{Primary loading} \times \text{Secondary offtake} \times \text{Stock norms} \times \text{Billing frequency})$

Collection = $f(\text{Outstanding aging} \times \text{Collection rate} \times \text{Credit discipline})$

When revenue at a geography declines, the contextual detail panel walks this chain: is the decline driven by reach erosion (fewer outlets billing), extraction decay (lower WSP per outlet), pipeline stuffing (primary loading without secondary offtake), or collection issues (outstanding growing, restricting fresh billing)? Each branch maps to specific drift rules and specific packaged screens.

KPI Registry

Full KPI list with formula, L2 source, causal position, and monitoring drift rule. Covers: ND%, WD%, GEO ECO%, extraction rate, WSP/outlet, UoS, lines/call, Sec: Pri ratio, DMS: Pri, OFR, MAU, PJP MAU, QAU, pipeline ratio, days stock, billing frequency, SALY (primary), SALY (secondary), SALM, outstanding >30d, collection rate, promo ROI, promo participation rate, demand-distribution gap, market share estimate, visit compliance %, productive calls %.

06 Action Layer

Most actions in trade distribution are conversations, not system triggers. The Command Centre's job is to make those conversations informed, contextual, and traceable — not to replace them.

The 80/20 Split

80%: Discussion Actions

The RSM sees a finding. Picks up the phone. Calls the ASM. "What's happening in that district?" This is how trade distribution actually works. The Command Centre serves this workflow by making findings shareable and by capturing the response.

Shareable finding cards. Every drift finding renders as a self-contained card: the metric, the decomposition, the map snippet, the distributor list. The card is exportable as an image for WhatsApp. The RSM sends it to a group with the question "what's going on here?" The ASM replies with a voice note, a photo from the market, a location pin for the distributor depot.

Org context graph. Responses — notes, voice transcripts, photos, location markers — get tagged back to the finding. Over time, this accumulates the tribal knowledge that traditional dashboards lose. When the same geography drifts again six months later, the previous context is attached. The finding is not just a number — it carries the organisational memory of what was discussed and what was done.

Distribution of interactive artefacts. Beyond WhatsApp cards, the Command Centre generates click-through reports for specific audiences — a QBR pack for the NSM, a territory review for the RSM, a beat health summary for the ASM. These are packaged screen exports with scope pre-set. They go out on a cadence (weekly, monthly) or on-demand when a finding warrants it.

WhatsApp as primary channel. Images of finding cards with spatial context. Click-through links to drill-down screens. Voice notes from field with ground context tagged back. Location markers for outlets, distributors, depots referenced in discussions. This is where 80% of trade action happens and the Command Centre meets users here.

20%: War Room Actions

These are the half-yearly or annual strategic exercises: beat redesign, expansion bursts, network restructuring, SD appointment waves. They are not triggered by a button on a screen. They are planned, deliberate, and have their own lifecycle.

Headless design. The relevant Clarynt module or Growth Agent runs the computation — beat optimisation, expansion targeting, SD appointment scoring. The Command Centre's role is to provide the input context (which geographies, which constraints, which thresholds) drawn from accumulated drift findings and org context.

Dissemination. The war room output — new beat books, target outlet lists, revised territory boundaries — gets pushed to the field via SFA, flat file export, or WhatsApp packs. The Command Centre tracks dissemination status: sent, received, acknowledged.

Program management. After dissemination, the Command Centre monitors adoption against the design. Are new beats being followed? Are target outlets being visited? Are expansion zones showing billing? This is where war room actions loop back into drift detection — DA-05 (Beat Productivity Drift) monitors whether the redesigned beats are performing to design intent.

Action Loop Flow

31. **Drift finding surfaces.** On landing screen, ranked by severity.
32. **Discussion action (80%).** User shares finding on WhatsApp, calls ASM, collects ground context. Notes and media tagged back to finding in org context graph.
33. **Finding status updates.** New → Viewed → Discussed → Resolved or Escalated. Escalated findings accumulate into the case for a war room exercise.
34. **War room action (20%).** When enough escalated findings concentrate in a geography or theme, the team convenes a war room. Input: accumulated findings + org context. Module/agent runs the design. Output disseminated to field. Adoption monitored via drift detection.
35. **L2 re-aggregates.** Next data refresh picks up field changes (new outlets billing, revised beat compliance, scheme adjustments). Drift detection re-runs. Findings resolve or persist.

Drift Finding Lifecycle

Status flow:
new → viewed → discussed (notes attached) → resolved
→ escalated → war_room_queued → actioned → monitoring

Table: finding_context (RDS Postgres)
context_id, finding_id, context_type (note|voice_transcript|photo|location|report),
content (text or S3 URI), author (user_id or "field_response"),
source_channel (whatsapp|call|in_app|sfa), created_at

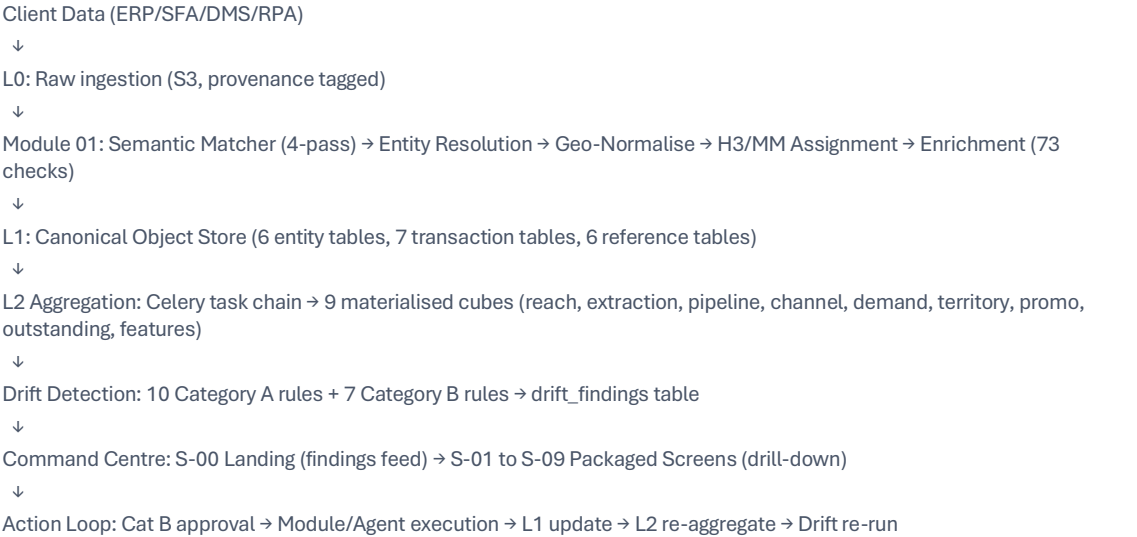
Table: war_room_exercises
exercise_id, title, geography_scope, theme (expansion|beat_redesign|scheme_revision|...),
linked_finding_ids[], module_used, agent_used,
status (planning|designing|disseminating|monitoring|closed),
output_artifacts[] (beat_books, target_lists, scheme_params),
dissemination_status (jsonb: {channel, sent_count, ack_count}),
created_at, closed_at

07 Technology Stack

L A Y E R	T E C H N O L O G Y	W H Y
L0 Storage	S3 / local filesystem	Raw data, immutable, audit trail
L1 Store	DuckDB + h3-duckdb + RDS Postgres	Columnar analytics + ACID state
L2 Aggregates	DuckDB materialised / Parquet	Pre-computed. Read-only for screens.
Drift Engine	Celery tasks. DuckDB queries against L2.	Post-aggregation. Dependency-aware.
State Store	RDS Postgres	drift_findings, user_screen_state, screen_config

API Layer	FastAPI	Stateless. Reads L2 + Postgres.
Frontend	React + Clarynt v4 tokens	Dumb renderer. Reads view specs.
Spatial	Leaflet (MVP) → deck.gl + MapLibre	Embedded, contextual.
Orchestration	Celery + Redis (EC2)	Shared with Growth Agent pipelines.

08 End-to-End Data Flow



End of Technical Architecture v1.4

Next: L2 aggregation SQL definitions, first 3 drift rules on synthetic data, S-00 and S-01 wireframe proof.